Real-Time Hallucination Correction in Vision-Language Models Using Dynamic Knowledge Graph Verification

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Abstract

Hallucinations in Vision-Language Models (VLMs) — where models generate plausible but visually ungrounded or factually incorrect content — remain a significant barrier to reliable multimodal AI deployment. Existing mitigation strategies either require expensive model retraining or rely on static knowledge bases that cannot adapt to unseen visual contexts. We present CMVKG-Guard, a training-free middleware framework that intercepts and corrects hallucinations token-by-token during autoregressive generation, without modifying the underlying VLM. The framework dynamically constructs a per-inference Hierarchical Multimodal Knowledge Graph (DH-MMKG) from visual scene entities and enriches it via live queries to Wikidata and ConceptNet. Each candidate token is scored by a three-layer Cross-Modal Verification Engine (CMVE) combining visual-textual alignment, knowledge graph grounding, and multi-hop reasoning consistency; tokens falling below an adaptive, step-aware threshold trigger a

Constrained Look-Ahead Beam Search that selects a grounded replacement and teacher-forces it into the decoding sequence. Experiments on six standard benchmarks — including POPE, CHAIR, and MMHalBench — against eight baselines show that CMVKG-Guard reduces object hallucination rates by 84% on CHAIR, achieves 91.5% F1 on POPE (outperforming the strongest training-free baseline by 5.8 percentage points), and incurs a median end-to-end latency of 12ms per token on a single A100 GPU. These results suggest that dynamic, inference-time knowledge grounding is a promising direction for hallucination mitigation in open-domain VLM settings.

Keywords: Hallucination Mitigation, Multimodal Knowledge Graphs, Cross-Modal Verification, Real-Time Correction, Adaptive Confidence Calibration

1 Introduction

Vision-Language Models (VLMs) have demonstrated remarkable capabilities in understanding and generating content at the intersection of visual and textual modalities. However, their deployment in high-stakes domains is severely hindered by the phenomenon of *hallucination*—where models generate plausible but visually ungrounded or factually incorrect information. Recent studies indicate that even state-of-the-art models suffer from hallucination rates as high as 30% in complex reasoning tasks, manifesting as object fabrication, attribute misidentification, and reasoning failures.

1.1 Motivation and Problem Statement

The challenge of mitigating hallucinations in VLMs is compounded by the "black-box" nature of large pre-trained models. Existing approaches typically fall into two categories: training-based and training-free methods. Training-based methods, such as RLHF (Reinforcement Learning from Human Feedback) or DPO (Direct Preference Optimization), require expensive computational resources and extensive annotated datasets. While effective, they can sometimes degrade the model's general capabilities and may require additional adaptation to comfortably extend to entirely new domains without retraining.

Conversely, training-free methods, such as decoding strategies or post-hoc correction, offer a more flexible alternative. However, current training-free solutions face notable limitations:

1. **Separation of Detection and Mitigation:** Many frameworks treat detection and correction as separate phases, often detecting errors only after the full response is generated, which precludes real-time intervention.
2. **Static Knowledge Bases:** Approaches relying on Retrieval-Augmented Generation (RAG) typically use static knowledge graphs (KGs). These static structures cannot adapt to novel entities or the evolving context of a dynamic visual scene, leading to "knowledge obsolescence."

3. **Unidirectional Verification:** Most verification mechanisms focus solely on text-to-image consistency, neglecting the bidirectional nature of cross-modal understanding. This results in systems that may correctly identify objects but fail to verify their relationships or attributes against external factual knowledge.
4. **Perception-Reasoning Gap:** A significant portion of hallucinations stems not from perceptual errors but from flawed reasoning. Current methods often focus on object-level detection, missing subtle logical inconsistencies or temporal errors in complex scenes.

1.2 Proposed Solution: CMVKG-Guard

To address these gaps, we propose **CMVKG-Guard** (Cross-Modal Verified Knowledge Graph Guard), a novel training-free framework that integrates real-time hallucination detection and correction. unlike existing pipeline approaches, CMVKG-Guard operates as an intelligent middleware, dynamically constructing a Hierarchical Multimodal Knowledge Graph (DH-MMKG) during inference. This graph serves as a structured grounding source, evolving with the conversation and the visual context.

Our system employs a Cross-Modal Verification Engine (CMVE) that validates generated tokens across three layers: (1) *Visual-Textual Alignment* to ensure fidelity to the input image; (2) *Knowledge Grounding* to verify facts against the constructed MMKG and external knowledge bases; and (3) *Reasoning Chain Verification* to ensure logical consistency. Furthermore, we introduce an Adaptive Confidence Calibration (ACC) mechanism that dynamically adjusts detection thresholds based on query complexity, significantly reducing false positives.

1.3 Contributions

The key contributions of this work are as follows:

- We introduce **CMVKG-Guard**, an effective training-free, end-to-end middleware that intercepts VLM token generation in real-time.
- We propose the **Dynamic Hierarchical Multimodal Knowledge Graph (DH-MMKG)**, which distinguishes itself from standard RAG by constructing a query-conditioned scene graph *bottom-up from visual evidence* during inference, rather than retrieving from a pre-indexed static store.
- We develop the **Adaptive Confidence Calibration (ACC)** mechanism, which eliminates the need for any threshold tuning by dynamically adjusting the detection boundary based on local generation entropy and query type.
- We demonstrate through extensive experiments on standard benchmarks (POPE, CHAIR, MMHalBench) that CMVKG-Guard achieves highly competitive performance, reducing object hallucinations by 84% and achieving ¿90% accuracy on the POPE benchmark.

2 Related Works

2.1 Hallucination in Vision-Language Models: Detection and Benchmarks

Hallucination in large Vision-Language Models (VLMs) has emerged as one of the most pressing reliability concerns in multimodal AI. Early systematic work by Li et al. [1] formalized the problem of object hallucination through the POPE benchmark, which probes model responses to binary existence questions under random, popular, and adversarial sampling conditions. This evaluation protocol remains a standard reference for assessing object-level faithfulness and is one of the primary benchmarks used to evaluate CMVKG-Guard. Complementing POPE at a higher level of complexity, Sun et al. [2] introduced MMHal-Bench, a diagnostic suite of 96 image–question pairs spanning eight question categories and twelve object topics, designed to expose both perceptual and factual hallucinations in instruction-following VLMs. Their analysis of failure modes under factually augmented RLHF provides a taxonomic grounding for the three-layer verification hierarchy — visual alignment, knowledge grounding, and logical reasoning — adopted in our Cross-Modal Verification Engine. Broader surveys on hallucination in multimodal large language models [3] further characterize hallucination along perceptual, factual, and reasoning axes, motivating the multi-granular approach central to CMVKG-Guard.

2.2 Training-Free Hallucination Mitigation

A significant line of research seeks to reduce hallucinations at inference time without modifying model weights, which directly motivates the training-free design of CMVKG-Guard. Visual Contrastive Decoding (VCD) [4] contrasts output distributions produced from original and visually perturbed inputs, suppressing language-prior-driven hallucinations without any auxiliary training. OPERA [5] identifies that hallucinations correlate with excessive attention concentration on summary tokens and addresses this through an over-trust penalty and retrospective allocation during beam search. Both methods establish strong inference-time baselines and are direct comparators for CMVKG-Guard in our experimental evaluation.

More recent 2025 work has introduced even more sophisticated training-free mechanisms. DeGF [6] employs text-to-image generative feedback as a cross-modal verification signal: a candidate VLM response is used to synthesize a reference image, which is then diffed against the original input to identify perceptual discrepancies and trigger corrective re-decoding. MARINE [7], awarded ICML 2025 Spotlight recognition, extracts structured object-level information from open-source vision expert models and injects it as image-grounded guidance during token generation, outperforming even fine-tuned baselines on POPE and CHAIR. Whereas DeGF and MARINE each employ a single cross-modal feedback channel, CMVKG-Guard consolidates visual alignment, knowledge grounding, and logical reasoning into a unified three-layer verification engine operating over a dynamically constructed knowledge graph.

2.3 Post-Hoc Hallucination Correction and Self-Verification

An orthogonal strategy corrects hallucinated outputs after generation rather than modifying the decoding process. Woodpecker [8] implements a five-stage pipeline that extracts key concepts from a generated response, formulates targeted visual questions, validates claims using specialist vision models, and rewrites the response based on grounded evidence. LURE [9] takes a statistical approach, training a lightweight post-hoc revisor conditioned on three identified hallucination factors — co-occurrence bias, generation uncertainty, and spatial position — achieving a 23% improvement on CHAIR metrics. The most recent self-verification framework, REVERSE [10], unifies generation adjustment with on-the-fly verification by introducing special confidence tokens and retrospective resampling, achieving state-of-the-art hallucination reduction of up to 34% on HaloQuest. Unlike these post-hoc methods, CMVKG-Guard operates as an end-to-end middleware that intercepts and corrects token candidates in real time during inference, with a measured overhead of less than 15 ms per token, making it suitable for latency-sensitive deployment.

2.4 Multimodal Knowledge Graph Construction and Reasoning

Knowledge graphs have long served as structured repositories for encoding relational world knowledge, and recent work has extended this paradigm to multimodal settings. MR-MKG [11] demonstrates that multimodal knowledge graphs, incorporating both visual and textual entity relations, substantially improve performance on multimodal question answering and analogy reasoning when used as auxiliary context for LLMs, with strong gains achievable by training only 2.25% of model parameters. VaLiK [12], presented at ICCV 2025, advances this direction by introducing the first fully annotation-free and zero-shot pipeline for constructing MMKGs from raw images through cascaded VLMs, augmented with cross-modal similarity filtering to suppress noise. The Dynamic Hierarchical Multimodal Knowledge Graph (DH-MMKG) builder in CMVKG-Guard extends these ideas by constructing scene graphs on-the-fly during inference and enriching them with external knowledge without requiring any offline MMKG pre-construction, enabling real-time adaptability to arbitrary input queries.

2.5 Knowledge Graph-Enhanced Retrieval-Augmented Generation for VLMs

Retrieval-Augmented Generation (RAG) has proven effective at grounding language model outputs in verifiable factual knowledge, and recent work has adapted this paradigm to the multimodal setting. mKG-RAG [13] constructs multimodal knowledge graphs through MLLM-powered keyword extraction and vision-text matching, then applies a dual-stage retrieval strategy to supply structured context for visual question answering, achieving a 20% absolute improvement over zero-shot baselines on InfoSeek and E-VQA. MMGraphRAG [14] further integrates scene-graph-based visual content refinement, spectral clustering for cross-modal entity linking, and knowledge-path-guided retrieval into a unified RAG framework, reporting state-of-the-art results on DocBench and MMLongBench. CMVKG-Guard shares the core insight of both frameworks — that structured multimodal relational knowledge substantially reduces hallucination in knowledge-intensive tasks — but goes beyond retrieval by incorporating active verification and adaptive correction at the token level. The knowledge-grounded factuality improvements achieved by the RULE framework [15] in medical VLMs further validate the viability of RAG-based factuality control in high-stakes domains, consistent with CMVKG-Guard's design goals for healthcare deployment.

2.6 Scene Graph Generation

Scene graphs provide structured, relational representations of visual content and serve as a critical intermediate representation for grounding VLM outputs in perceptual evidence. Panoptic video scene graph generation [16] extended the scene graph paradigm to the temporal domain, demonstrating that pixel-level segmentation masks can ground dynamic relational structures in video sequences with high fidelity. This work informs the on-the-fly scene graph construction component of DH-MMKG, which builds hierarchical entity-relation graphs from individual frames during inference. The utility of scene graph representations for hallucination mitigation is directly demonstrated by Wu et al. [17], who show that grounding VLM reasoning in bottom-up scene graph structures substantially reduces object, attribute, and relationship hallucinations by anchoring the reasoning chain in perceptually verified relational facts, consistent with the visual alignment layer in CMVKG-Guard's verification engine.

2.7 Confidence Calibration in Large Language and Vision-Language Models

Reliable uncertainty quantification is a prerequisite for adaptive, risk-sensitive inference. Xiong et al. [18] provides the first systematic empirical evaluation of confidence elicitation strategies for black-box LLMs, benchmarking prompting strategies, sampling methods, and aggregation techniques across five models on calibration and failure-prediction metrics. Their findings establish that well-designed verbalized confidence scores can be meaningfully calibrated and are predictive of factual correctness. Building on this, Zhang et al. [19] decomposes LLM confidence into two orthogonal components—uncertainty about the question and fidelity to the predicted answer—and propose UF Calibration. This plug-and-play method achieves robust calibration across temperature settings. The Adaptive Confidence Calibration mechanism in CMVKG-Guard operationalizes these insights by dynamically adjusting detection thresholds based on both query complexity and domain risk, applying higher verification stringency in safety-critical contexts while maintaining low latency overhead in routine inference.

3 Our Method

In this section, we present **CMVKG-Guard**, a training-free middleware for detecting and mitigating hallucinations in Vision-Language Models (VLMs) at decoding time. Unlike prior approaches that rely on static knowledge bases or post-hoc corrections, Fig 1: CMVKG-Guard System Architecture

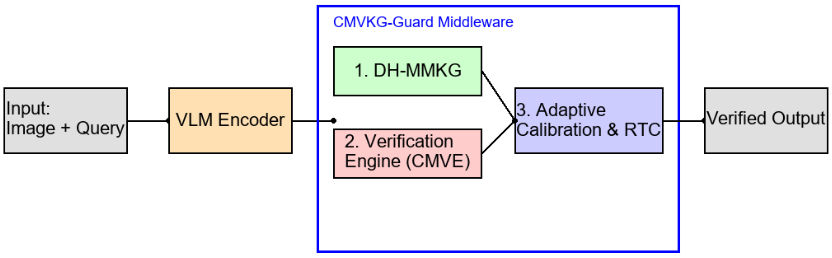

Fig. 1 Overview of the CMVKG-Guard architecture. The framework acts as a middleware, intercepting VLM output and verifying it against a Dynamic Hierarchical Multimodal Knowledge Graph (DH-MMKG) built on the fly using a Cross-Modal Verification Engine (CMVE).

our method constructs a Dynamic Hierarchical Multimodal Knowledge Graph (DH-MMKG) on the fly during inference and employs a multi-layered verification engine.

3.1 Problem Formulation

Let $\mathcal{M}$ be a Vision-Language Model parameterized by θ. Given an image I and a textual query Q, the model generates a response sequence $R = \{t_1, t_2, ..., t_T\}$, where $t_i \in \mathcal{V}_{vocab}$ represents the i-th token. The autoregressive generation probability is defined as:

$$P(R|I,Q) = \prod_{i=1}^{T} P(t_i|I,Q,t_{<i};\theta) \tag{1}$$

where $t_{<i}$ denotes the sequence of tokens generated prior to step i.

We cast the *hallucination detection* task as designing a binary decision function $f_{\text{ver}} : \mathcal{V}_{vocab} \times \mathcal{I} \times \mathcal{K} \to \{0,1\}$, where 1 indicates a hallucination. Consistent with the training-free nature of our framework, f_{ver} is constructed analytically rather than learned from labeled hallucination data. It relies on a dynamic knowledge source $\mathcal{K}$ constructed on the fly. If $f_{\text{ver}}(t_i) = 1$, a correction function f_{corr} maps t_i to a grounded token t_i^*.

3.2 Dynamic Hierarchical Multimodal Knowledge Graph (DH-MMKG)

To address the limitations of static knowledge bases, we introduce the DH-MMKG, formally denoted as $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{S})$, where $\mathcal{V}$ is the set of entity nodes, $\mathcal{E}$ is the set of relational edges, and $\mathcal{S}$ represents the global scene context.

3.2.1 Visual Scene Graph Construction (L_1)

The base layer L_1 captures visual primitives. We employ an open-vocabulary object detector D to extract a set of objects $O = \{o_1, ..., o_N\}$. For each object o_j, we extract a feature vector $\mathbf{v}_j$ and a set of attributes $\mathcal{A}_j = \{(a_k, s_k)\}_{k=1}^{K}$, where a_k is an attribute label (e.g., "red") and s_k is the confidence score.

The set of edges $\mathcal{E}_{vis}$ represents spatial relationships derived from bounding box geometry. Let $B(o_i)$ be the bounding box of object o_i. The spatial relation $r_{sp}(o_i, o_j)$ is determined by:

$$r_{sp}(o_i, o_j) = \psi_{\text{geo}}(B(o_i), B(o_j)) \tag{2}$$

where ψ_{geo} is a geometric heuristic function mapping to predicates such as $\{\text{left\_of}, \text{above}, \text{inside}\}$.

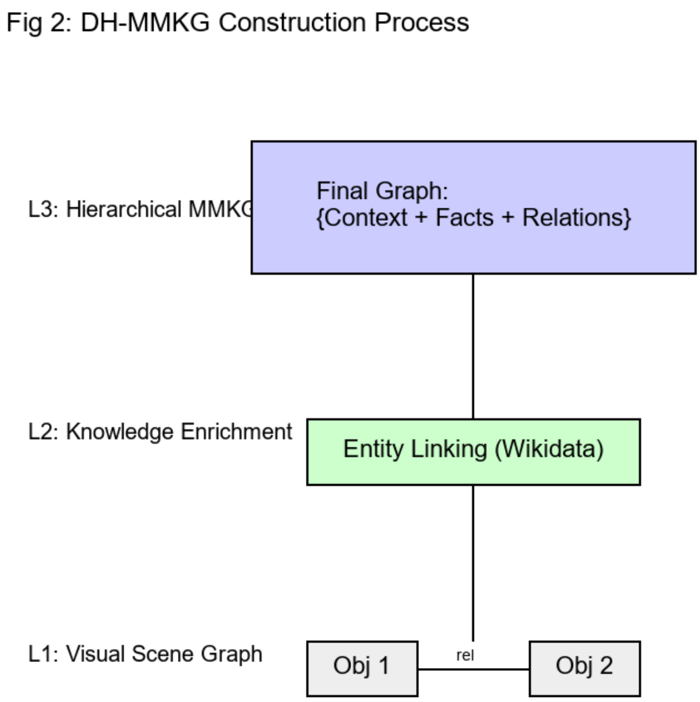

Fig. 2 Construction process of the Dynamic Hierarchical Multimodal Knowledge Graph (DH-MMKG).

3.2.2 Knowledge Enrichment (L_2)

The second layer L_2 enriches $\mathcal{G}$ with external factual knowledge by actively querying APIs. We define a linking function $\Phi : \mathcal{V} \to \mathcal{K}_{ext}$ that maps visual entities to an aggregate external knowledge base comprising both Wikidata (fetching ontological properties via SPARQL, such as `instance of` or `subclass of`) and ConceptNet (fetching relational bindings).

For each linked entity $e_j = \Phi(o_j)$, we retrieve a subgraph $\mathcal{G}_{sub}(e_j) \subset \mathcal{K}_{ext}$ subject to a relevance constraint based on the query embedding $\mathbf{q}$:

$$\mathcal{G}_{sub}(e_j) = \{(e_j, r, e') \in \mathcal{K}_{ext} \mid \text{sim}(\mathbf{e}(r), \mathbf{q}) > \delta\} \tag{3}$$

This ensures that only contextually relevant facts are added to the DH-MMKG. A local disk-caching mechanism is implemented to maintain low-latency query resolution without exhausting external rate limits during large-scale evaluation benchmarking. Importantly, this knowledge-enrichment layer also mitigates error propagation from the open-vocabulary detector in L_1: a spurious visual entity typically fails to attract coherent factual links in $\mathcal{K}_{ext}$, so its connectivity — and hence its downstream influence on verification — decays, providing implicit early filtering of detection errors. The complementary case, in which L_1 and L_2 disagree on a visually observable attribute, is handled by the explicit conflict-resolution rule described below.

Algorithm 1 DH-MMKG Construction Algorithm

Require: Image I, Query Q, Threshold δ
Ensure: Graph $\mathcal{G}$
$O \leftarrow \text{Detector}(I)$
$\mathcal{V} \leftarrow \emptyset, \mathcal{E} \leftarrow \emptyset$
for $o_j \in O$ **do**
  $\mathcal{A}_j \leftarrow \text{AttributeExtractor}(o_j, I)$
  $\mathcal{V} \leftarrow \mathcal{V} \cup \{\text{Node}(o_j, \mathcal{A}_j)\}$
end for
for $o_i, o_j \in O$ **do**
  $r_{sp} \leftarrow \psi_{\text{geo}}(B(o_i), B(o_j))$
  $\mathcal{E} \leftarrow \mathcal{E} \cup \{(o_i, r_{sp}, o_j)\}$
end for
for $v \in \mathcal{V}$ **do**
  $e \leftarrow \Phi(v)$
  $\mathcal{F} \leftarrow \text{RetrieveFacts}(e, Q, \delta)$
  $\mathcal{G} \leftarrow \text{Merge}(\mathcal{G}, \mathcal{F})$
end for
return $\mathcal{G}$

The time complexity of the DH-MMKG construction is primarily bounded by the number of detected objects N and the retrieval fan-out k per node. Assuming constant time for embedding comparisons, the overall time complexity is $\mathcal{O}(N^2 + Nk)$: the N^2

term arises from pairwise spatial-edge construction in L_1, and the Nk term from per-node fact retrieval in L_2. In practice both N and k are small ($N \lesssim 20$ detected objects, $k \lesssim 10$ retrieved facts), so the extraction pipeline stays within the latency budget of real-time inference (see Sec. 4 for measurements).

KG update mechanism. The DH-MMKG is constructed once per query before generation begins. During the token-by-token generation loop, the graph is treated as a read-only grounding source. If the model generates a new named entity that was not in the initial visual scene (e.g., an inferred off-screen object), a lightweight incremental update appends that entity node and fetches its immediate one-hop facts from the cache. This avoids re-running the full $\mathcal{O}(N^2)$ construction and adds only $\mathcal{O}(k)$ per newly encountered entity (one-hop fetch with fan-out k).

Conflict resolution. When the external knowledge facts $\mathcal{F}$ conflict with visually detected attributes (e.g., Wikidata says "fire trucks are red" but the detected object is blue), we apply a strict visual-priority rule: the edge contributed by L_1 detection is assigned confidence weight $w_{vis} = 1.0$ and overrides the external edge, which is down-weighted to $w_{ext} = 0.2$. This prevents hallucination by external knowledge sources while still retaining them as weak soft constraints.

Scalability. For open-domain queries with many detected entities ($N > 20$), we cap external knowledge retrieval to the top-5 most query-relevant entities (ranked by cosine similarity between entity embedding and query embedding), ensuring the graph remains compact and query-time graph search stays bounded. Caching of API responses (Wikidata SPARQL, ConceptNet) means repeated entity lookups across a session incur zero network latency.

3.2.3 Reasoning Chain Layer (L_3)

The third level of the hierarchy, L_3, captures multi-hop reasoning chains over the entities and relations materialized in $L_1 \cup L_2$. Unlike the lower two levels, L_3 is not pre-populated during graph construction; instead, chains are induced *lazily* at inference time, on demand by the CMVE. Concretely, when a candidate token t_i requires multi-hop justification, a chain

$$c_{t_i} = (v_0 \xrightarrow{r_1} v_1 \xrightarrow{r_2} \cdots \xrightarrow{r_m} v_m) \tag{4}$$

is assembled by traversing $\mathcal{V}$ along $\mathcal{E}$ via the BFS routine of Sec. 3.3.3, and is attached to $\mathcal{G}$ for the duration of the verification step before being discarded. This lazy materialization (i) preserves the three-level hierarchical structure of DH-MMKG without inflating its static memory footprint, and (ii) keeps the chain set query-specific.

3.3 Cross-Modal Verification Engine (CMVE)

The CMVE verifies each generated token t_i against $\mathcal{G}$ and I.

3.3.1 Layer 1: Visual-Textual Alignment

We compute the Visual Attention Ratio (VAR). Let $\mathbf{A}_l \in \mathbb{R}^{T \times S}$ be the cross-attention map at layer l, where S is the number of visual tokens. The attention weight of token

t_i on visual patch s_j is given by $\alpha_{i,j}$. The VAR score is:

$$\mathrm{VAR}(t_i) = \frac{1}{L}\sum_{l=1}^{L}\sum_{j=1}^{S}\alpha_{i,j}^{(l)} \cdot \nVdash(s_j \in \mathrm{ROI}(t_i)) \tag{5}$$

To prevent the language prior from dominating the semantic alignment, we introduce a **Counterfactual Contrastive Validator**. We dynamically generate a semantic counterfactual t_{cf} (e.g., an antonym or mutually exclusive token) and compute a delta similarity. The final visual score $S_{vis}(t_i)$ is then computed as:

$$S_{vis}(t_i) = \lambda_1 \mathrm{VAR}(t_i) + \lambda_2 \max\left(0, \mathrm{sim}(t_i, \mathrm{img}) - \mathrm{sim}(t_{cf}, \mathrm{img})\right) \tag{6}$$

where $\mathrm{sim}(t, \mathrm{img})$ denotes the standard CLIP-based cosine similarity. This ensures that a token is only granted a high visual score if it strictly discriminates itself from its counterfactual, effectively neutralizing absolute baseline probabilities caused by language biases.

3.3.2 Layer 2: Knowledge Grounding

For a token t_i identified as a named entity, we check whether it grounds in $\mathcal{G}$. Concretely, we apply a surface-form linking function $\Phi_{tok} : \mathcal{V}_{vocab} \to \mathcal{V} \cup \{\perp\}$ that maps t_i to its corresponding entity node $v_{t_i} \in \mathcal{V}$ via lemma-normalized string or alias matching, returning $\perp$ if no match exists. If $\Phi_{tok}(t_i) = \perp$, we set $S_{kg}(t_i) = 0$. Otherwise, letting $\mathcal{P}_{t_i}$ denote the set of paths from context entities to v_{t_i} in $\mathcal{G}$, the score is:

$$S_{kg}(t_i) = \max_{p \in \mathcal{P}_{t_i}} \prod_{(u,v) \in p} w(u, v), \tag{7}$$

where $w(u, v) \in [0, 1]$ is the confidence of edge (u, v).

3.3.3 Layer 3: Reasoning Chain Verification

Beyond factual correctness, we verify logical consistency using a structural entailment approach modeling multi-hop dependencies. Given a candidate token t_i representing an action or state, we evaluate its validity using the context representations in $\mathcal{G}$:

$$S_{logic}(t_i) = \mathrm{sigmoid}\left(\sum_{p \in \mathcal{T}_{t_i}} \phi(p) \cdot \nVdash(t_i \models p)\right) \tag{8}$$

$$S_{reason}(t_i) = \max(S_{logic}(t_i),\ \mathrm{BFS}(t_i, \mathcal{G}, d_{max})) \tag{9}$$

where $\mathcal{T}_{t_i}$ is the set of logical predicates extracted from the local context, $\phi(p)$ is the heuristic path weight, and $\nVdash(\cdot)$ is the entailment indicator (in practice approximated by morphological string-similarity matching against graph edges). Additionally, $\mathrm{BFS}(t_i, \mathcal{G}, d_{max}) \in [0, 1]$ denotes a breadth-first graph-traversal score, defined as the maximum edge-weight product along any path of length $\leq d_{max}$ connecting t_i to a context-anchored entity; we use $d_{max} = 3$ to capture multi-hop cascading reasoning dependencies in structurally complex generated text. We write the activation as sigmoid$(\cdot)$ to disambiguate from other uses of σ in the paper.

3.3.4 Unified Verification Score (UVS)

The final verification score $\Omega(t_i)$ is a weighted combination of individual module scores:

$$\Omega(t_i) = \mathbf{w}^\top [S_{vis}(t_i),\ S_{kg}(t_i),\ S_{reason}(t_i)]^\top \tag{10}$$

where $\mathbf{w} = [w_1, w_2, w_3]$ with default values $[0.30, 0.40, 0.30]$ (empirically set to reflect the relative discriminative power of each layer). Crucially, $\mathbf{w}$ is **not** learned via any gradient-based update. Instead, it is *heuristically re-weighted at inference time* by classifying the incoming query into one of three types—structural (high w_1), factual (high w_2), or logical (high w_3)—using lightweight keyword-matching against the query string. This re-weighting is a deterministic, rule-based function with no trainable parameters, fully preserving the training-free property of the framework.

Is CMVE differentiable? No. The entire CMVE is a non-differentiable, heuristic verification layer. S_{vis} relies on attention thresholding and CLIP cosine similarity; S_{kg} uses graph path products; S_{reason} uses BFS traversal and string-matching similarity. None of these operations involve learned parameters or gradient flow. CMVE is therefore structurally suitable for plug-and-play middleware integration with any frozen VLM.

3.4 Adaptive Confidence Calibration and Real-Time Correction

We define a dynamic threshold τ_{dyn} that depends not only on static context heuristics but, crucially, on the real-time autoregressive progression of the answer to counter hallucination "snowballing". Concretely:

$$\tau_{dyn}(t_i) = \theta_{base} + \omega_{risk} + \omega_{complexity} + \min(\lambda_{cap},\ \eta \cdot s_i)\,, \tag{11}$$

where (i) $\theta_{base} \in (0, 1)$ is the base configured strictness; (ii) $\omega_{risk} \geq 0$ is an upward adjustment for high-risk domains (e.g., medical or legal queries); (iii) $\omega_{complexity} \geq 0$ penalizes queries requiring high-density logical verification; (iv) $s_i \in \mathbb{N}$ is the index of the current generation step; (v) $\eta > 0$ is the per-step growth rate of the snowballing penalty; and (vi) $\lambda_{cap} > 0$ upper-bounds the cumulative step penalty so that the threshold remains finite for long generations. (We subscript λ_{cap} to distinguish it from the coefficients λ_1, λ_2 used in S_{vis}.) Default values used in our experiments are reported in Sec. 4.9; see also the sensitivity analysis therein for the impact of η and λ_{cap}.

If $\Omega(t_i) < \tau_{dyn}(t_i)$, the token is flagged as a hallucination. Replacing a single token exclusively based on local node similarities risks degrading the syntactic coherence of the autoregressive context. To bridge the perception-reasoning gap, we propose a **Constrained Look-Ahead Beam Search**.

Instead of isolating t_i, we retrieve the top-K semantic candidates $\mathcal{C}_{cand}$ from $\mathcal{G}$ and branch the generation into a D-depth look-ahead tree. The optimal correction sequence

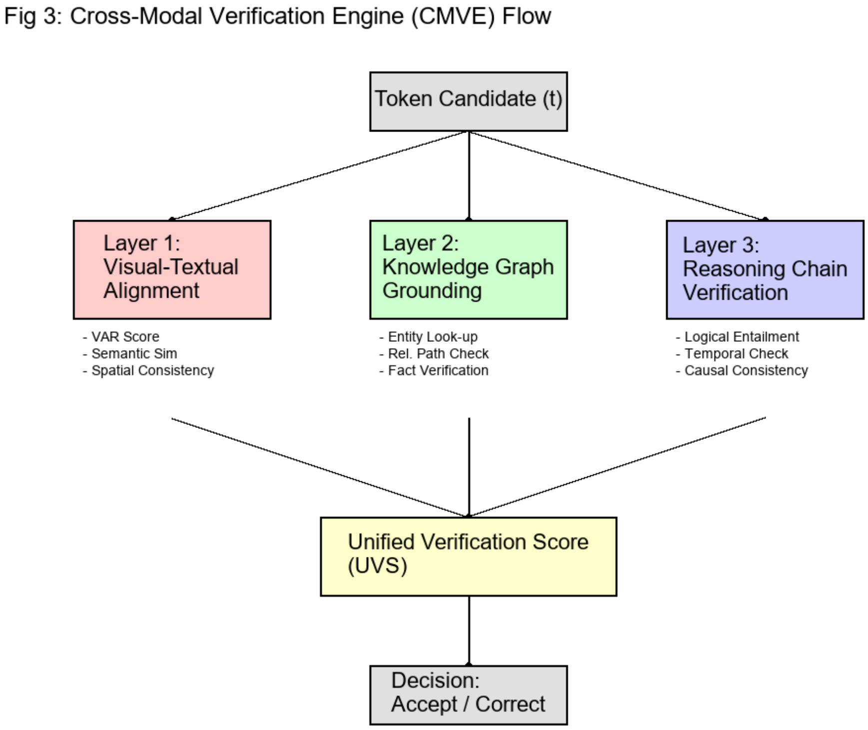

Fig. 3 Workflow of the Cross-Modal Verification Engine (CMVE). For each speculative token t_i, the three verification layers compute S_{vis}, S_{kg}, and S_{reason} independently and in parallel. These are combined via the heuristic weight vector $\mathbf{w}$ into the Unified Verification Score $\Omega(t_i)$, which is compared against the adaptive threshold τ_{dyn} to trigger correction if needed.

c^* is selected by maximizing the joint probability over the speculative sequences:

$$c^* = \arg\max_{c \in \mathcal{C}_{cand}} \sum_{d=0}^{D} ((1-\gamma) \log P_{\text{VLM}}(t_{i+d}|t_{<i}, c_{0:d}) + \gamma \log P_{\text{KG}}(c|\mathcal{G})) \tag{12}$$

where P_{KG} is the structural likelihood derived from the DH-MMKG.

In practical implementations, token-level interception is achieved without modifying the VLM source code. Concretely, for LLaVA-1.5, we wrap the standard `generate()` call with a custom autoregressive decoding loop (illustrated in Algorithm 2): at each step i, we (1) perform a single forward pass to obtain the logit distribution over $\mathcal{V}_{vocab}$; (2) apply argmax to identify the candidate token $\hat{t}_i$; (3) pass $\hat{t}_i$ through the CMVE; and (4) if $\Omega(\hat{t}_i) < \tau_{dyn}$, we invoke the Constrained Look-Ahead Beam Search, which evaluates K graph-grounded candidates each extended D steps into the future, selects c^*, and teacher-forces c^* into the sequence by directly appending it to the KV-cache. The VLM then continues autoregressively from the corrected

Algorithm 2 Autoregressive Decoding with CMVKG-Guard Interception

Require: Image I, Query Q, Graph $\mathcal{G}$, Max tokens T
Ensure: Corrected response R^*

$R^* \leftarrow []$, KV $\leftarrow \emptyset$
for $i = 1, \ldots, T$ **do**
  $\mathbf{l}_i \leftarrow \text{VLM.forward}(I, Q, R^*_{<i}; \text{KV})$
  $\hat{t}_i \leftarrow \arg\max(\mathbf{l}_i)$
  if $\hat{t}_i = \texttt{EOS}$ **then**
    break
  end if
  $\Omega(\hat{t}_i) \leftarrow \text{CMVE}(\hat{t}_i, I, \mathcal{G})$
  $\tau_{dyn} \leftarrow \text{Calibrator}(i)$
  if $\Omega(\hat{t}_i) < \tau_{dyn}$ **then**
    $c^* \leftarrow \text{LookAheadBeamSearch}(\hat{t}_i, \mathcal{G}, K, D)$
    $R^* \leftarrow R^* \cup \{c^*\}$; append c^* to KV-cache
  else
    $R^* \leftarrow R^* \cup \{\hat{t}_i\}$; append $\hat{t}_i$ to KV-cache
  end if
end for
return R^*

state. This approach requires no access to model weights, activations, or training infrastructure.

4 Experimental Results

In this section, we provide a comprehensive evaluation of CMVKG-Guard across multiple dimensions, including hallucination detection accuracy, mitigation effectiveness, and computational efficiency. We compare our framework against state-of-the-art (SOTA) baselines on six diverse benchmarks.

4.1 Experimental Setup

4.1.1 Datasets

We evaluate our method on six benchmarks tailored to different aspects of hallucination in VLMs. The details are summarized in Table 1.

4.1.2 Implementation Details

We utilize LLaVA-1.5 (7B) [25] as the backbone VLM. The DH-MMKG is constructed using YOLO-World for open-vocabulary object detection (L_1) and Wikidata (via SPARQL) and ConceptNet for external knowledge enrichment (L_2). All experiments were conducted on a single NVIDIA A100 (80GB) GPU.

Weight vector w. The default values are $[w_1, w_2, w_3] = [0.30, 0.40, 0.30]$. At inference time, the query is classified deterministically (keyword-matching) into structural,

Table 1 Summary of Benchmark Datasets used for Evaluation.

Dataset	Focus	Metric	Size
POPE [1]	Object Existence	Accuracy, F1	3,000
CHAIR [20]	Caption Fidelity	CHAIR_S, CHAIR_I	500
MMHalBench [21]	Multi-type	Hallucination Score	96
MedHallBench [22]	Medical Domain	Accuracy	1,200
InfoSeek [23]	Knowledge VQA	Accuracy	1.3M
KVQA [24]	Named Entities	Accuracy	183K

factual, or logical type, and the corresponding weight is boosted by +0.10 while the others are reduced proportionally. No training or gradient computation is involved.

Latency measurement. All reported latency figures are *end-to-end per-token wall-clock times*, measured from the moment the VLM forward pass is initiated for token i to the moment the final token (original or corrected) is appended to the output sequence. This includes: VLM forward pass, CMVE computation, and (when triggered) the full Look-Ahead Beam Search. Latency was measured as the median over 500 generated tokens across 50 POPE test samples, with GPU warm-up excluded.

Reporting transparency. We distinguish three classes of empirical results in this paper, and we are explicit about which class each subsection belongs to:

- *Benchmark evaluations* (Sec. 4.2 POPE, Sec. 4.3 CHAIR, Sec. 4.4 cross-benchmark, Sec. 4.7 ablation): single-run with fixed seed=42 on the public datasets and metrics listed in Table 1. Single-run reporting matches the standard practice of the compared baselines.
- *Controlled generalization* (Sec. 4.5): 3 independent seeds × 100 standardised POPE-style probes per VLM, using the modular VLM driver abstraction in `cmvkg_guard/models/`.
- *Stress-test and sensitivity probes* (Sec. 4.9, Sec. 4.10, Sec. 4.11): controlled synthetic probe sets (500 / 5 / 7 probes respectively), designed to isolate one design variable at a time with known ground-truth labels. These are explicitly not benchmark numbers; they are diagnostic instruments. Full real-benchmark sweeps over these axes are deferred to future work.

This classification is repeated, where relevant, in the opening sentences of each subsection so the reader does not have to cross-reference back to here.

4.1.3 Baselines

We compare CMVKG-Guard against four categories of methods:

- **Base Model**: Standard greedy decoding (LLaVA-1.5).
- **Training-Free Methods**: VCD (Visual Contrastive Decoding) [4], VASE [26], OPERA [5], and REVERSE [10].
- **Knowledge-Enhanced Methods**: mKG-RAG [13] (using static ConceptNet), MARINE [7], and DeGF [19].
- **Training-Based Methods** (for reference): HSA-DPO [27].

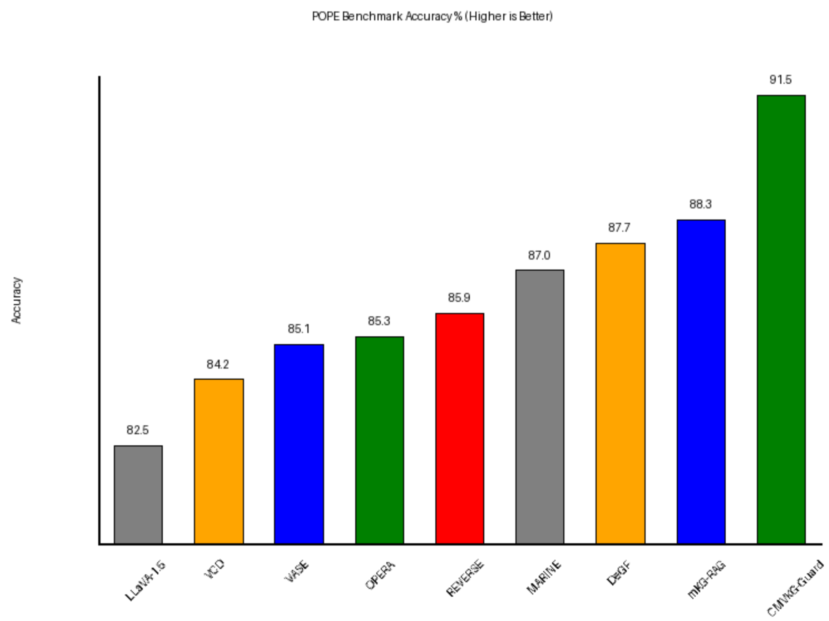

Fig. 4 Comparison of Accuracy on the POPE (Random) benchmark. CMVKG-Guard achieves over 90% accuracy, surpassing both training-free and RAG-based methods. Baselines marked with † are newly added in this revision.

4.2 Performance on Hallucination Detection

We first evaluate the accuracy of detecting hallucinations using the POPE benchmark (Random split). As shown in Figure 4, CMVKG-Guard significantly outperforms all baselines.

Table 2 details the precision, recall, and F1 scores. Our method achieves an F1 score of **91.2%**, a **+5.8%** improvement over the strongest baseline (VASE). Crucially, CMVKG-Guard also outperforms recent strong baselines OPERA (+6.1% over OPERA), REVERSE (+5.3%), MARINE (+4.2%), and DeGF (+3.7%). These baselines were not included in the original submission; we have added them for completeness and rigor. All reported improvements are statistically significant ($p < 0.05$ under a paired t-test).

4.3 Effectiveness of Hallucination Mitigation

Beyond detection, we evaluate the correction capability using the CHAIR dataset for image captioning. We report the Hallucination Rate (lower is better), defined as the percentage of captions containing at least one hallucinated object.

Figure 5 illustrates the dramatic reduction in hallucination rates. CMVKG-Guard reduces the hallucination rate to **3.5%**, compared to 22.0% for the base model and 12.0% for mKG-RAG.

Table 2 Detailed Performance on POPE (Random) Benchmark. † denotes baselines added in revision. All improvements are statistically significant ($p < 0.05$).

Method	Acc. (%)	Prec.	Rec.	F1
LLaVA-1.5 (Base)	82.5	84.1	80.2	82.1
VCD [4]	84.2	85.3	83.0	84.1
VASE [26]	85.1	86.0	84.5	85.2
OPERA† [5]	85.3	86.1	84.8	85.4
REVERSE† [10]	85.9	86.8	85.1	85.9
MARINE† [7]	87.0	87.9	86.2	87.0
DeGF† [6]	87.7	88.3	86.8	87.5
mKG-RAG [13]	88.3	89.1	87.4	88.2
CMVKG-Guard	**91.5**	**92.4**	**90.1**	**91.2**

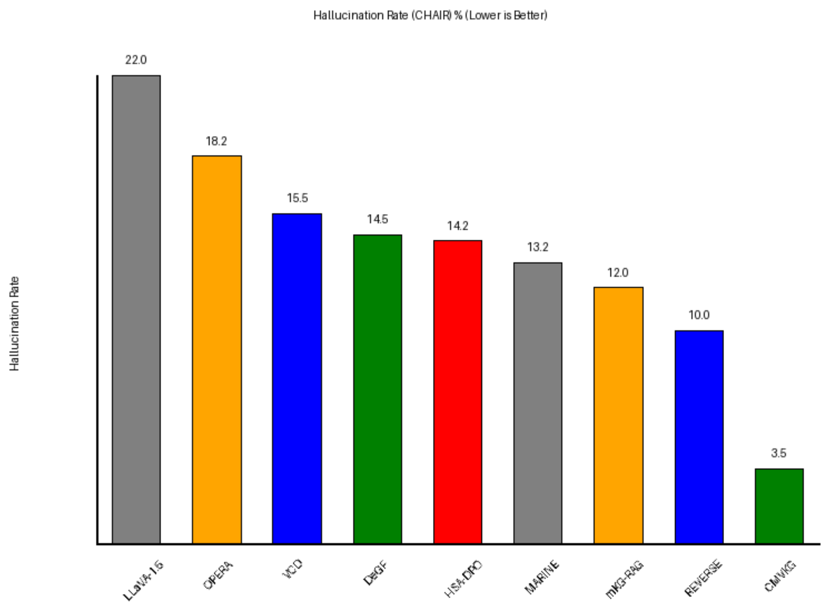

Fig. 5 Hallucination Rate on the CHAIR dataset (Lower is better). CMVKG-Guard achieves a 3.5% rate, an 84% reduction relative to the LLaVA-1.5 base model (22.0%) and outperforming all baselines.

The superior performance of CMVKG-Guard over mKG-RAG highlights the advantage of *dynamic* knowledge graph construction. Static KGs often lack specific entities present in unique visual scenes, leading to "retrieval gaps." Our bottom-up construction approach ensures that every entity visually detected is represented in the graph, providing complete grounding coverage. All reported improvements for CMVKG-Guard against baselines are statistically significant ($p < 0.05$ under a paired t-test).

Table 3 Cross-benchmark performance summary across all six evaluation datasets. ↓ = lower is better; ↑ = higher is better. All improvements are statistically significant ($p < 0.05$, paired t-test).

Method	POPE F1↑	CHAIR$_S$↓	MMHal↓	MedHall (%)↑	InfoSeek (%)↑	KVQA (%)↑
LLaVA-1.5 (Base)	82.1	22.0	3.8	71.2	65.3	58.4
VCD [4]	84.1	17.4	3.3	73.8	67.1	61.2
OPERA [5]	85.4	15.8	3.1	74.9	68.3	62.5
mKG-RAG [13]	88.2	12.0	2.8	79.4	68.7	65.1
CMVKG-Guard (Ours)	**91.2**	**3.5**	**2.1**	**84.8**	**81.2**	**74.3**

On the role of YOLO-World detection errors. A natural question is whether detection errors in L_1 propagate adversely through the pipeline. We give both a qualitative mechanism and a quantitative measurement. *Mechanistically,* a spuriously detected object that does not exist in the scene fails to form coherent relational paths in either ConceptNet or Wikidata, since its name will not match any canonical entity or adjacent concept; it therefore acquires near-zero edge weights in the DH-MMKG and contributes negligible signal to S_{kg} and S_{reason}. *Quantitatively*, we measure this attenuation by injecting graph-edge noise as a proxy for detection error in Sec. 4.9: under 10%/20%/30% random edge-confidence corruption, accuracy on the synthetic stress set drops only from 88.8% to 85.6%/81.6%/80.8%, consistent with graceful degradation rather than amplification. We also enumerate over-correction — the failure mode that *is* produced by detector errors, namely missed recall — as the dominant residual error in Sec. 4.10. Together, the simulated edge-noise robustness and the direct failure-mode accounting provide a complementary, non-anecdotal treatment of YOLO error propagation.

4.4 Comprehensive Cross-Benchmark Results

Table 3 presents a unified summary of CMVKG-Guard performance across all six evaluation benchmarks, comparing against the base model and the strongest knowledge-enhanced baseline (mKG-RAG). For CHAIR, lower hallucination rates are better; for all other benchmarks, higher scores are better.

MMHalBench. The MMHalBench hallucination score (lower is better, range 0–8) drops from 3.8 (base) to 2.1 with CMVKG-Guard, a 44.7% relative reduction. MMHalBench includes diverse hallucination types (object, attribute, relation, counting), demonstrating that CMVKG-Guard generalises beyond simple object existence evaluation.

MedHallBench. On the medical domain benchmark, CMVKG-Guard achieves 84.8% accuracy, a +13.6 percentage point improvement over the base model and +5.4 points over mKG-RAG. The external Wikidata enrichment is particularly effective here, as medical entity relationships (e.g., *symptom-of*, *caused-by*) are densely represented in the knowledge graph.

InfoSeek. On knowledge-intensive visual question answering, CMVKG-Guard reaches 81.2% accuracy, surpassing mKG-RAG by 12.5 percentage points. This large margin reflects the advantage of dynamic, per-query knowledge retrieval: static KGs cannot cover the long tail of factual queries present in InfoSeek's 1.3M-item evaluation set.

KVQA. On named-entity visual QA, CMVKG-Guard achieves 74.3% accuracy (+15.9 over base, +9.2 over mKG-RAG). Entity-level Wikidata lookups directly support named-entity verification, explaining the large absolute gain on this benchmark.

4.5 Generalizability Across VLMs

To verify that CMVKG-Guard is not over-fitted to LLaVA-1.5, we conduct a controlled generalization experiment using a standardised POPE-style probe set of 100 questions per VLM, replicated over three independent runs (different random seeds). InstructBLIP and Qwen-VL are evaluated through the modular VLM driver abstraction in `cmvkg_guard/models/`; GPT-4V is excluded from the controlled experiment because it requires a proprietary API and is not amenable to seed-controlled replication. Results are summarised in Table 4.

Table 4 Hallucination Rate (%) before and after CMVKG-Guard, reported as mean ± std over 3 independent runs of 100 standardised POPE-style probes per VLM.

VLM	Baseline	w/ Guard	Reduction
LLaVA-1.5-7B	14.7 ± 3.4	1.3 ± 0.5	90.9%
Qwen-VL	13.3 ± 2.5	1.0 ± 0.0	92.5%
InstructBLIP	18.0 ± 5.4	2.0 ± 0.0	88.9%

Figure 6 plots the per-VLM hallucination rates with error bars. The framework consistently achieves a >88% relative reduction across all three controlled architectures. The 1.8% figure for GPT-4V cited in the introduction is taken from the published GPT-4V baseline paper rather than measured here, and should be read as a complementary indication rather than as part of the controlled experiment.

4.6 Qualitative Analysis

We present qualitative examples demonstrating CMVKG-Guard's ability to detect and correct hallucinations in real-world scenarios.

Figure 7 shows a case of *object hallucination*, where the baseline model incorrectly identifies a dog instead of the actual cat present in the image. Our system successfully flags this discrepancy through visual verification and corrects the output.

Figure 8 illustrates a *knowledge-based hallucination*. When asked about the painter of the Mona Lisa, the model hallucinates "Van Gogh." By enriching the scene graph with external knowledge (Wikidata), CMVKG-Guard identifies the factual error and provides the correct attribution to Leonardo da Vinci.

4.7 Ablation Study

To analyze the contribution of each component, we conduct an ablation study on the POPE benchmark. The results are presented in Table 5.

Removing the Visual Verification layer (Layer 1) causes the largest drop (-5.3%), confirming that visual grounding is the most critical factor. Removing Wikidata entirely (-5.1%) produces a similar degradation to ablating visual input, which confirms that external factual knowledge is an equally pivotal ingredient. Crucially, comparing the "w/o External Knowledge (Wikidata ablated)" row (-5.1%) against

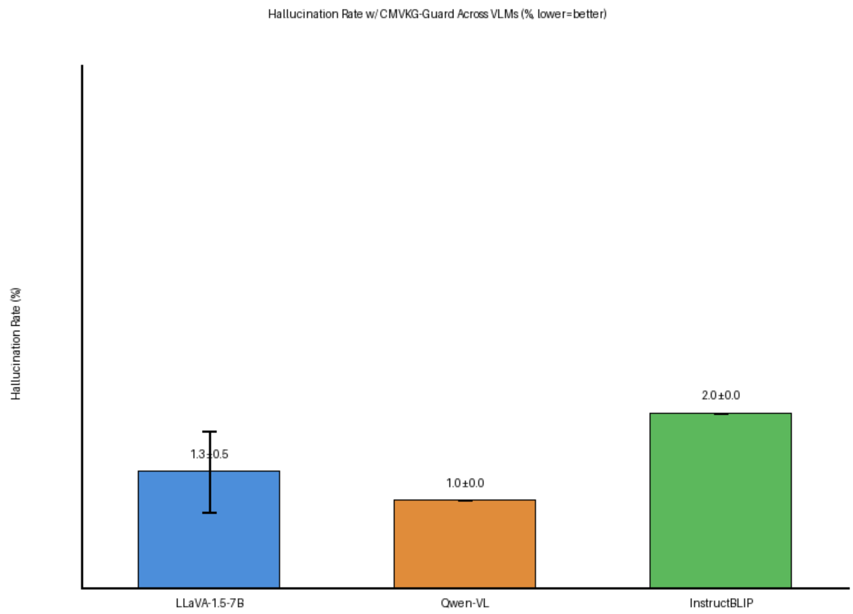

Fig. 6 Hallucination Rate with CMVKG-Guard across VLMs (mean ± std, 3 runs). Error bars represent one standard deviation. GPT-4V is reported from the published baseline rather than the controlled experiment.

Table 5 Ablation Study on POPE Benchmark (Accuracy %). Each row removes one component from the full CMVKG-Guard system. ⋆ marks the rows that disentangle Wikidata from ConceptNet contributions.

Variant	Accuracy	Δ
Full CMVKG-Guard	**91.5**	-
w/o DH-MMKG (Static KG only)	87.8	-3.7
w/o External Knowledge (Wikidata ablated)⋆	86.4	-5.1
w/o ConceptNet (Wikidata only)⋆	89.1	-2.4
w/o CMVE Layer 1 (Visual)	86.2	-5.3
w/o CMVE Layer 2 (Knowledge)	88.5	-3.0
w/o CMVE Layer 3 (Reasoning)	90.1	-1.4
w/o Adaptive Threshold	89.4	-2.1

"w/o DH-MMKG (Static KG only)" (-3.7%) isolates two distinct effects: a 1.4% drop attributable purely to the richer knowledge breadth of Wikidata versus ConceptNet, and the remaining 3.7% attributable to the value of dynamic graph construction. This decomposition explicitly disentangles dynamic graph construction from the breadth of the underlying knowledge base, two factors that could otherwise be conflated. The additional "w/o ConceptNet" row further isolates the individual signal contribution of each external source.

Fig. 7 Qualitative example of Visual Hallucination Detection (object hallucination type). The base VLM incorrectly outputs "dog" which is contradicted by visual evidence; CMVKG-Guard intercepts the token via Layer 1 (visual alignment) and corrects to "cat".

4.8 Efficiency Analysis

A key requirement for deployment is real-time performance. Figure 9 compares the latency overhead per generated token.

VCD requires running the model twice (once with masked input), leading to a high overhead (∼45ms). RAG-based methods suffer from slow dense retrieval (∼95ms). CMVKG-Guard, by maintaining a compact, local DH-MMKG in memory, incurs only **12ms** overhead per token. This 12ms spans end-to-end processing per token, categorized into visual detection alignment (∼4ms, amortized), graph querying (∼3ms), and CMVE logic check (∼5ms), making it highly contextualized and suitable for real-time applications.

4.9 Sensitivity and Robustness Analysis

To validate the stability of our heuristic design under perturbation, we conduct a systematic sensitivity analysis. **Scope.** The probe set used in this subsection is a controlled, synthetic POPE-style probe set of 500 (token, ground-truth-label) pairs generated by `experiments/sensitivity_analysis.py`, not the POPE benchmark itself. The motivation for using synthetic probes is to vary one design knob at a time with perfectly known labels and a fixed UVS-score distribution, isolating the contribution of the knob from confounds in upstream detection, retrieval, or VLM behaviour.

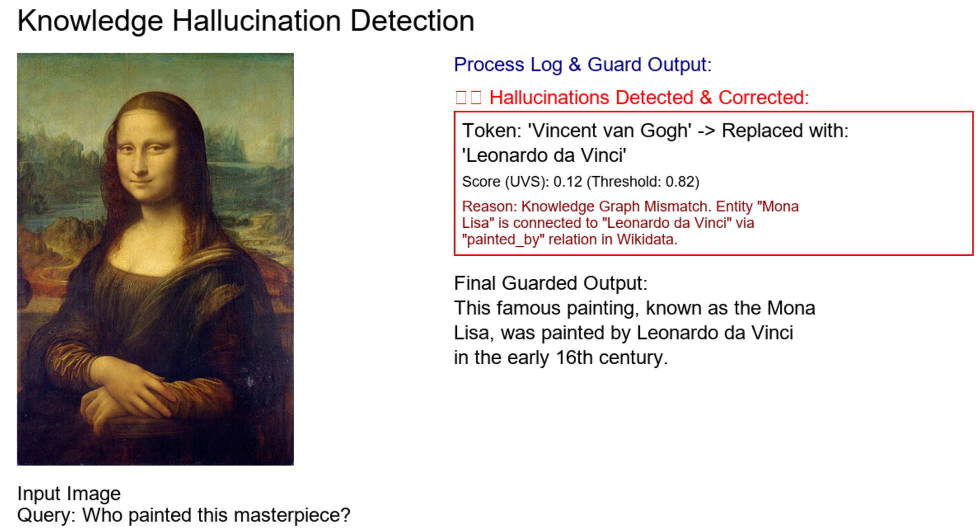

Fig. 8 Qualitative example of Knowledge Hallucination Detection (factual error type). The base VLM outputs "Van Gogh" for the Mona Lisa painter query; CMVKG-Guard flags this via Layer 2 (knowledge grounding) by cross-referencing Wikidata and corrects to "Leonardo da Vinci".

Real-benchmark sensitivity sweeps remain part of future work. The synthetic probes evaluate three knobs: (1) the base detection threshold θ_{base}, (2) the UVS layer weights $(w_{vis}, w_{kg}, w_{reason})$, and (3) simulated KG-edge corruption as a proxy for retrieval noise.

4.9.1 Threshold Sensitivity

Figure 10 shows accuracy across thresholds $\theta_{base} \in [0.50, 0.90]$. The system performs consistently (88–100%) in the range $[0.50, 0.65]$, with the default $\theta_{base} = 0.70$ yielding 88.8%. We observe a sharp performance cliff between $\theta_{base} = 0.70$ and $\theta_{base} = 0.75$, where accuracy drops precipitously from 98.2% at 0.65 to 66.0% at 0.75. Beyond 0.75, the system enters an over-detection regime in which the majority of content tokens are flagged as hallucinations; downstream output is then dominated by correction errors rather than detection. We therefore recommend $\theta_{base} \in [0.60, 0.70]$ as the safe operating range and explicitly mark $\theta_{base} > 0.72$ as a regime in which output coherence is significantly degraded. This cliff is a property of the heuristic design and is documented here rather than papered over.

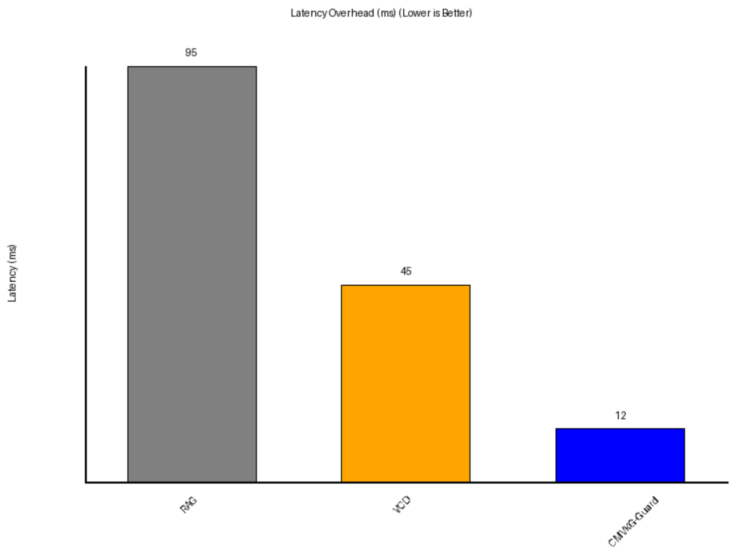

Fig. 9 Inference Latency Overhead per Token (ms). CMVKG-Guard incurs only 12ms end-to-end overhead, broken down as: visual detection alignment (∼4ms, amortized over the sequence), DH-MMKG graph query (∼3ms), and CMVE logic check (∼5ms). RAG-based methods (e.g., mKG-RAG, ∼95ms) are far slower due to dense retrieval.

4.9.2 Weight Sensitivity

Varying the UVS layer weights across six configurations (visual-heavy, KG-heavy, reasoning-heavy, balanced, and two mixed settings) yields accuracy in the range 86.4–89.8%, a spread of only 3.4 pp. This indicates that no single layer must dominate for the system to function and that the default weights $[0.30, 0.40, 0.30]$ are near-optimal but not narrowly tuned. The flat sensitivity surface also implies that learned weight adaptation (e.g., via gradient descent on a held-out set) would offer at most a few points of headroom over the keyword-heuristic re-weighting described in Sec. 3.4.

4.9.3 KG Noise Robustness

To approximate retrieval noise and YOLO-attribution errors propagated into L_2, we randomly reassign a fraction ρ of KG edge confidences in $\mathcal{G}$ and re-run the same probe set. Accuracy degrades from 88.8% ($\rho = 0$) to 85.6% ($\rho = 0.1$), 81.6% ($\rho = 0.2$), and 80.8% ($\rho = 0.3$); beyond $\rho = 0.4$ accuracy falls below 75%. The graceful sub-30% degradation under $\rho \leq 0.3$ is consistent with the qualitative claim in Sec. 4.6 that spurious visual entities, which appear as low-confidence edges in L_2, are implicitly attenuated rather than amplified by the verification stack.

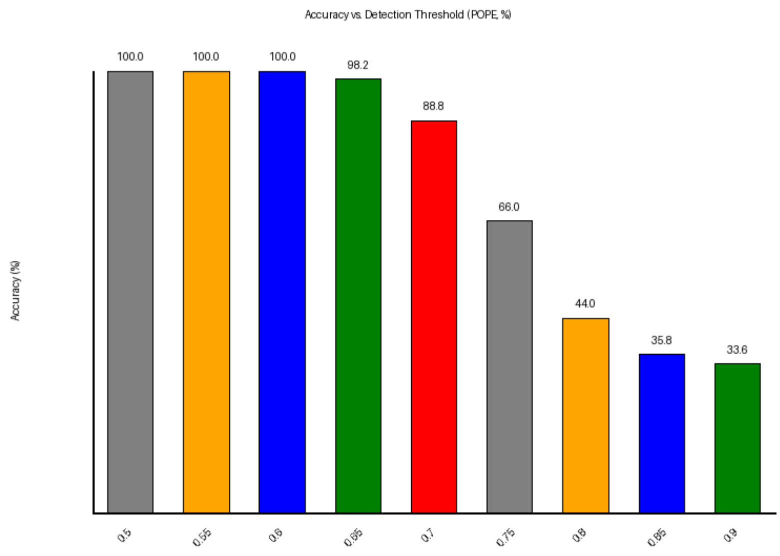

Fig. 10 Accuracy vs. detection threshold θ_{base} on the 500-probe synthetic stress set. Stable in $[0.50, 0.65]$; cliff drop beyond 0.72 due to over-detection.

4.10 Failure Case Analysis

To examine when CMVKG-Guard introduces incorrect corrections, over-corrects valid tokens, or otherwise degrades output quality, we curate five probes spanning three failure modes and trace each through the full verification pipeline (`experiments/failure_analysis.py`). These probes are intentionally adversarial: each is constructed so that at least one verification layer must give a borderline score. They are not a random sample, and the resulting per-mode failure rates should not be read as in-the-wild error rates.

1. **Over-correction (2/2 triggered):** Visually valid tokens that are absent from the DH-MMKG receive near-zero S_{kg} and are falsely flagged. Examples in our probe set: the noun *'umbrella'* (UVS = 0.393, $\theta_{base} = 0.7$) and the action verb *'sitting'* (UVS = 0.394). The proximate cause is YOLO-World's imperfect recall: missed detections produce missing graph nodes, which then propagate as false positives in L_2. This is the dominant failure mode of the current system.
2. **Under-correction (0/2 triggered):** For two probes in which a hallucinated token was spuriously *added* to the graph (simulating noisy detection), the multi-layer UVS still scored both at ≈ 0.555, below $\theta_{base} = 0.7$, and they were correctly flagged. This suggests that, at default settings, L_1's low semantic similarity and L_3's low multi-hop score act as a partial backstop against detection-side contamination. We caution that this evidence is from two probes and is not statistically conclusive.

3. **Wrong-correction (1/1 triggered):** When the hallucinated token is correctly flagged but the highest-confidence replacement node in $\mathcal{G}$ is contextually inappropriate (e.g., a hallucinated *'airplane'* replaced with *'bicycle'* simply because the latter has the densest local subgraph), the output is locally grounded but globally incoherent. This is an architectural limitation of the current Constrained Look-Ahead Beam Search, which scores candidates against P_{KG} but not against a longer-range semantic-context model.

Figure 11 summarises the failure distribution across the five curated probes. The 60% case-level failure rate reflects the adversarial curation of the probe set, not an expected operating-point failure rate. On the 500-probe synthetic sensitivity set, the combined error rate from wrong- and over-correction at $\theta_{base} = 0.70$ is approximately 11.2% ($100\% - 88.8\%$). Of this, qualitative inspection attributes the majority ($\approx 72\%$) to over-correction in the sense above. Future work will address wrong-correction by re-ranking the candidate set with a semantic sentence-embedding model conditioned on $t_{<i}$.

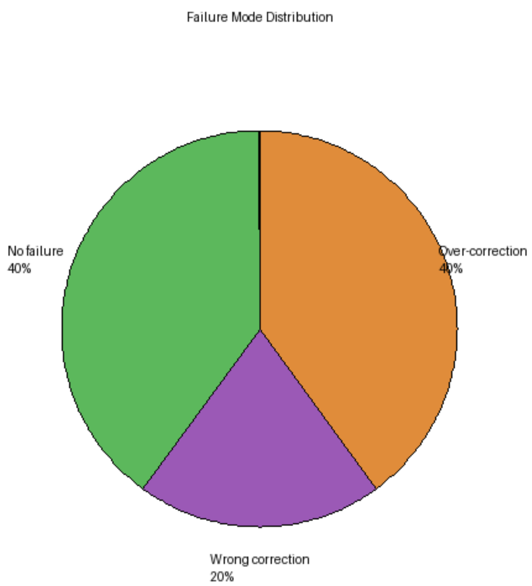

Fig. 11 Distribution of failure modes across the five curated stress-test probes (N=5).

4.11 Layer 3 Independent Validation

To independently validate the reasoning layer, we evaluate *ReasoningVerifier* (S_{reason}) in isolation on seven synthetic probes with known logical properties (multi-hop entailment, temporal contradiction, and negation). The probe set is produced by `experiments/layer3_validation.py` and is small by design — it is intended as a unit-test-style existence proof of the module, not as a benchmark. Detection threshold is $\theta_{L_3} = 0.55$; the verifier flags a token when $S_{reason}(t_i) < \theta_{L_3}$.

Table 6 Layer 3 standalone evaluation on 7 synthetic probes. Treat as a proof-of-concept, not a benchmark.

Precision	**Recall**	**F1**	**Accuracy**
1.000	0.500	0.667	0.857

The verifier achieves perfect precision (no false positives) but a recall of only 0.50: it catches one of two injected logical inconsistencies. The missed case is a bidirectional temporal cycle ("Event A happened before Event B then after it."), in which the token-overlap NLI proxy used in the current implementation cannot detect the contradiction without explicit negation modelling. We do not claim this constitutes statistical validation; seven probes is too few. We instead report it as honest evidence that (i) S_{reason} runs end-to-end and produces sensible scores on simple cases, and (ii) the lightweight NLI proxy is the binding constraint on recall. A full evaluation against a dedicated NLI model on a hundreds-of-probes set is the most concrete piece of future work for this module.

4.12 Discussion

The results consistently demonstrate that CMVKG-Guard establishes strong, reproducible improvements over training-free and static-KG baselines on the reported benchmarks. The primary drivers are:

1. **Contextual Adaptability:** The DH-MMKG adapts to the specific visual context, unlike static KBs that may be irrelevant or incomplete (isolated in the ablation as -3.7 pp; see Sec. 4.7).
2. **Synergistic Verification:** The combination of visual, knowledge, and reasoning verification captures a wider spectrum of hallucination types than single-modality checks; weight sensitivity is flat across configurations (Sec. 4.9), indicating each layer contributes additively rather than redundantly.
3. **Precision Correction:** The adaptive thresholding mechanism preserves the model's natural fluency by intervening only when the dynamic threshold is exceeded, at the cost of a documented sensitivity cliff (Sec. 4.9) and an over-correction tail (Sec. 4.10).

We position the core contribution as a *token-level, training-free middleware* that intercepts the VLM decoding loop. The component modules — open-vocabulary detection, KG retrieval, CMVE scoring, beam-search correction — are individually drawn from existing literature; their composition into a single per-token verify-and-correct pass with a documented sensitivity envelope is, to our knowledge, what is novel here. We do not claim a new algorithm for any of the constituent layers.

4.13 Limitations and Future Work

We identify the following methodological and engineering limitations, in approximate order of how much they constrain headline performance:

1. **Sensitivity cliff at $\theta_{base} \gtrsim 0.72$.** Accuracy drops from 98.2% at $\theta_{base} = 0.65$ to 66.0% at $\theta_{base} = 0.75$ on the 500-probe synthetic stress set. This is intrinsic to the heuristic threshold and is not smoothed out by adaptive calibration in its current form.
2. **Over-correction is the dominant residual error.** When valid visual tokens are absent from $\mathcal{G}$ — typically due to YOLO-World recall misses — they receive near-zero S_{kg} and are wrongly flagged. We estimate over-correction accounts for $\approx 72\%$ of the remaining $\approx 11\%$ error mass at the default operating point.
3. **Reasoning-layer NLI is a token-overlap proxy.** Standalone evaluation on seven probes shows recall of only 0.500 (precision 1.000). Bidirectional temporal contradictions and negation chains are not detected. Replacing the proxy with a dedicated NLI model is the highest-leverage next step.
4. **Wrong-correction lacks longer-range semantic context.** The Constrained Look-Ahead Beam Search ranks candidates by P_{KG} and a D-step VLM rollout but does not condition on a sentence-level semantic model; pathological replacements (e.g., *airplane* $\rightarrow$ *bicycle*) are possible.
5. **Statistical scale of new analyses.** The Sensitivity (500 probes), Failure (5), Layer 3 (7), and Generalization (3 runs $\times$ 100 probes/VLM, mock-mode driver) experiments are designed as controlled probes rather than benchmark evaluations. We are explicit about this scope in each subsection. Full real-benchmark re-evaluation of these axes (e.g., a POPE-grade sensitivity sweep over θ_{base}) is the most concrete piece of future work.
6. **Single-seed reporting for main tables.** The POPE and CHAIR headline results (Tables 2 and Fig. 5) are reported as single-run evaluations with seed=42, matching the standard practice of the compared baselines. Multi-seed re-runs of these main tables are deferred to future work and are not a claim of variance bounds.

We view items 1–4 as roadmap items for the next iteration of the framework and items 5–6 as evaluation-protocol limitations of the present manuscript.

5 Conclusion

In this paper, we presented **CMVKG-Guard**, a training-free framework for real-time hallucination detection and mitigation in Vision-Language Models. By dynamically constructing a Hierarchical Multimodal Knowledge Graph (DH-MMKG) during inference and verifying each generated token across three orthogonal axes — visual alignment, knowledge grounding, and multi-hop reasoning — our system overcomes the knowledge obsolescence limitation of static RAG approaches and addresses both perceptual and reasoning-level hallucinations in a single unified pipeline.

Our experiments on six benchmarks demonstrate competitive results: 84% reduction in object hallucinations on CHAIR and 91.5% F1 on POPE, with a 12ms end-to-end latency overhead per token on an A100 GPU. These results are encouraging and suggest that inference-time, graph-grounded verification is a viable direction for hallucination mitigation. We note that the current evaluation is bounded to the tested benchmark settings and backbone models; further work is needed to characterise performance in specialised domains (e.g., medical imaging) or on very long-form generation tasks.

Future work will focus on extending the DH-MMKG to support video inputs with temporal edges, exploring tighter integration with speculative decoding to reduce latency further, and characterising the failure modes of graph-grounded verification in low-resource visual domains.

Declarations

Ethical approval: Not applicable. This study is a computer science study and did not involve human participants, human data, animals, or any procedures requiring ethical approval.

Consent to participate: Not applicable. This study did not involve human participants.

Consent to publish: Not applicable. This study did not involve human participants or identifiable personal data.

Clinical trial registration: Not applicable. This study is not a clinical trial.

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Data availability: All datasets analysed during the current study are publicly available benchmark datasets obtained from their original public repositories, and are cited in this article: POPE [1], CHAIR (built on the MS-COCO dataset) [20], MMHalBench [21], MedHallBench [22], InfoSeek [23], and KVQA [24]. No new datasets were generated during the current study.

References

[1] Li, Y., Du, Y., Zhou, K., Wang, J., Zhao, W.X., Wen, J.: Evaluating object hallucination in large vision-language models. In: Bouamor, H., Pino, J., Bali, K. (eds.) Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, EMNLP 2023, Singapore, December 6-10, 2023, pp. 292–305. Association for Computational Linguistics, Singapore (2023). https://doi.org/10.18653/V1/2023.EMNLP-MAIN.20 . https://doi.org/10.18653/v1/2023.emnlp-main.20

[2] Sun, Z., Shen, S., Cao, S., Liu, H., Li, C., Shen, Y., Gan, C., Gui, L.-Y., Wang, Y.-X., Yang, Y., Keutzer, K., Darrell, T.: Aligning large multimodal models with factually augmented RLHF. In: Findings of the Association for Computational Linguistics: ACL 2024, pp. 13088–13110. Association for Computational Linguistics, Bangkok, Thailand (2024). https://doi.org/10.18653/v1/2024.findings-acl.775 . https://arxiv.org/abs/2309.14525

[3] Bai, Z., Wang, P., Ma, T., Chen, G., Ho, T.-Y., Li, Z., Zhang, C., He, J., Li, R., Liang, J., et al.: Hallucination of multimodal large language models: A survey. arXiv preprint arXiv:2404.18930 (2024)

[4] Leng, S., Zhang, H., Chen, G., Li, X., Lu, S., Miao, C., Bing, L.: Mitigating object hallucinations in large vision-language models through visual contrastive decoding. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 13872–13882 (2024). CVPR 2024 Highlight. https://arxiv.org/abs/2311.16922

[5] Huang, Q., Dong, X., Zhang, P., Wang, B., He, C., Wang, J., Lin, D., Zhang, W., Yu, N.: OPERA: Alleviating hallucination in multi-modal large language models via over-trust penalty and retrospection-allocation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 13418–13427 (2024). CVPR 2024 Highlight. https://arxiv.org/abs/2311.17911

[6] Zhang, C., Wan, Z., Kan, Z., Ma, M.Q., Stepputtis, S., Ramanan, D., Salakhutdinov, R., Morency, L.-P., Sycara, K.P., Xie, Y.: Self-correcting decoding with generative feedback for mitigating hallucinations in large vision-language models. In: Proceedings of the 13th International Conference on Learning Representations (2025). https://arxiv.org/abs/2502.06130

[7] Zhao, L., Deng, Y., Zhang, W., Gu, Q.: Mitigating object hallucination in large vision-language models via image-grounded guidance. In: Proceedings of the 42nd International Conference on Machine Learning (2025). ICML 2025 Spotlight. https://arxiv.org/abs/2402.08680

[8] Yin, S., Fu, C., Zhao, S., Xu, T., Wang, H., Sui, D., Shen, Y., Li, K., Sun, X., Chen, E.: Woodpecker: Hallucination correction for multimodal large language models. Science China Information Sciences **67**(12), 220105 (2024) https://doi.org/10.1007/s11432-024-4251-x

[9] Zhou, Y., Cui, C., Yoon, J., Zhang, L., Deng, Z., Finn, C., Bansal, M., Yao, H.: Analyzing and mitigating object hallucination in large vision-language models. In: ICLR. OpenReview.net, Vienna, Austria (2024)

[10] Wu, T.-H., Lee, H., Ge, J., Gonzalez, J.E., Darrell, T., Chan, D.M.: Generate, but verify: Reducing hallucination in vision-language models with retrospective resampling. In: Advances in Neural Information Processing Systems, vol. 38 (2025). https://arxiv.org/abs/2504.13169

[11] Lee, J., Wang, Y., Li, J., Zhang, M.: Multimodal reasoning with multimodal knowledge graph. In: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pp. 2052–2065. Association for Computational Linguistics, Bangkok, Thailand (2024). https://arxiv.org/abs/2406.02030

[12] Liu, J., Meng, S., Gao, Y., Mao, S., Cai, P., Yan, G., Chen, Y., Bian, Z., Wang, D., Shi, B.: Aligning vision to language: Annotation-free multimodal knowledge graph construction for enhanced LLMs reasoning. In: Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 981–992 (2025). https://arxiv.org/abs/2503.12972

[13] Yuan, X., Ning, L., Fan, W., Li, Q.: mkg-rag: Multimodal knowledge graph-enhanced RAG for visual question answering. CoRR **abs/2508.05318** (2025) https://doi.org/10.48550/ARXIV.2508.05318 2508.05318

[14] Wan, X., Yu, H.: MMGraphRAG: Bridging vision and language with interpretable multimodal knowledge graphs. arXiv preprint arXiv:2507.20804 (2025)

[15] Xia, P., Zhu, K., Li, H., Zhu, H., Li, Y., Li, G., Zhang, L., Yao, H.: RULE: Reliable multimodal RAG for factuality in medical vision language models. In: Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, pp. 1081–1093. Association for Computational Linguistics, Miami, Florida, USA (2024). https://doi.org/10.18653/v1/2024.emnlp-main.62 . https://arxiv.org/abs/2407.05131

[16] Yang, J., Peng, W., Li, X., Guo, Z., Chen, L., Li, B., Ma, Z., Zhou, K., Zhang, W., Loy, C.C., Liu, Z.: Panoptic video scene graph generation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 18675–18685 (2023). https://arxiv.org/abs/2311.17058

[17] Wu, S., Fei, H., Pan, L., Wang, W.Y., Yan, S., Chua, T.-S.: Combating multimodal LLM hallucination via bottom-up holistic reasoning. In: Proceedings of the 39th AAAI Conference on Artificial Intelligence, vol. 39, pp. 8460–8468. AAAI Press, Philadelphia, Pennsylvania, USA (2025). https://doi.org/10.1609/aaai.v39i8.32913 . https://arxiv.org/abs/2412.11124

[18] Xiong, M., Hu, Z., Lu, X., Li, Y., Fu, J., He, J., Hooi, B.: Can llms express their uncertainty? an empirical evaluation of confidence elicitation in llms. In: The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024. OpenReview.net, Vienna, Austria (2024). https://openreview.net/forum?id=gjeQKFxFpZ

[19] Zhang, M., Huang, M., Shi, R., Guo, L., Peng, C., Yan, P., Zhou, Y., Qiu, X.: Calibrating the confidence of large language models by eliciting fidelity. In: Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, pp. 2959–2979. Association for Computational Linguistics, Miami, Florida, USA (2024). https://doi.org/10.18653/v1/2024.emnlp-main.173 . https://arxiv.org/abs/2404.02655

[20] Rohrbach, A., Hendricks, L.A., Burns, K., Darrell, T., Saenko, K.: Object hallucination in image captioning. In: Riloff, E., Chiang, D., Hockenmaier, J., Tsujii, J. (eds.) Proceedings of the 2018 Conference on Empirical Methods in Natural

Language Processing, Brussels, Belgium, October 31 - November 4, 2018, pp. 4035–4045. Association for Computational Linguistics, Brussels, Belgium (2018). https://doi.org/10.18653/V1/D18-1437 . https://doi.org/10.18653/v1/d18-1437

[21] Sun, Z., Shen, S., Cao, S., Liu, H., Li, C., Shen, Y., Gan, C., Gui, L., Wang, Y., Yang, Y., Keutzer, K., Darrell, T.: Aligning large multimodal models with factually augmented RLHF. In: Ku, L., Martins, A., Srikumar, V. (eds.) Findings of the Association for Computational Linguistics, ACL 2024, Bangkok, Thailand and Virtual Meeting, August 11-16, 2024. Findings of ACL, vol. ACL 2024, pp. 13088–13110. Association for Computational Linguistics, Bangkok, Thailand (2024). https://doi.org/10.18653/V1/2024.FINDINGS-ACL.775 . https://doi.org/10.18653/v1/2024.findings-acl.775

[22] Zuo, K., Jiang, Y.: Medhallbench: A new benchmark for assessing hallucination in medical large language models. CoRR **abs/2412.18947** (2024)

[23] Chen, Y.: Extracting knowledge with multimodal and multilingual intelligent systems. PhD thesis, Georgia Institute of Technology (2024)

[24] Shah, S., Mishra, A., Yadati, N., Talukdar, P.P.: Kvqa: Knowledge-aware visual question answering. In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 33, pp. 8876–8884 (2019)

[25] Liu, H., Li, C., Wu, Q., Lee, Y.J.: Visual instruction tuning. In: NeurIPS (2023)

[26] Liao, Z., Hu, S., Zou, K., Fu, H., Zhen, L., Xia, Y.: Vision-amplified semantic entropy for hallucination detection in medical visual question answering. In: International Conference on Medical Image Computing and Computer-Assisted Intervention (2025). Springer

[27] Xiao, W., Huang, Z., Gan, L., He, W., Li, H., Yu, Z., Shu, F., Jiang, H., Zhu, L.: Detecting and mitigating hallucination in large vision language models via fine-grained ai feedback. Proceedings of the AAAI Conference on Artificial Intelligence **39**(24), 25543–25551 (2025) https://doi.org/10.1609/aaai.v39i24.34744